

Creating and Deploying Production-Ready Agentic Systems

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What we will be talking about

- 1. The LLM as a System Component**
- 2. Tool-Based Reasoning using ReAct**
- 3. LLMs in Code Development**

Section 1

The LLM as a System Component

LLMs are not just task-specific tools

- So far, we have used LLMs as specialized tools for specific tasks.
- While useful, these tasks can generally be accomplished by **simpler** and more **scalable** models (with some performance loss).
- Where LLMs differ, is that they are **autonomous**, which allows for **previously impossible** use-cases.

When you are holding a hammer...

- Suppose you are given the following task, with appropriate resources from management.

Use case

Find all reviews that concern competitors of our product, and tell us what percentage of them mention our product.

... you treat everything as if it were a nail

The easy solution

We instruct an LLM to retrieve reviews via RAG. We isolate the reviews using prompting, then instruct the LLM to find whether our product is mentioned.

Result

Code is deployed **within the day**. Management is billed **50,000€** in a week.

But maybe we can do better

The system solution

We use a traditional scrapper to find relevant reviews. For each page, we instruct the LLM to extract the reviews. We then run `reviews.str.contains(product_name).sum()`.

Result

Code is deployed within **two days**. Management is billed **20€** in a week.

Decomposing the problem

Our problem could be decomposed in three parts

1. Discovery: Find pages containing reviews.
2. Filtering: Find the HTML sections containing the reviews.
3. Analysis: Find which reviews reference the product.

Which of these must necessarily be solved with LLMs?
Which can be solved with traditional ways? What trade-offs are we making in the latter case?

Section 2

Tool-Based Reasoning using ReAct

What is ReAct?

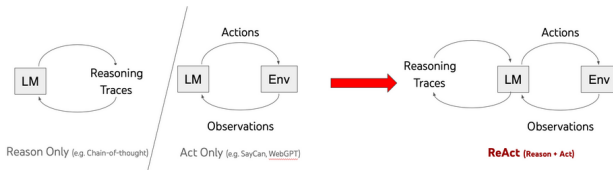


Figure: <https://arxiv.org/abs/2210.03629>

Definition

ReAct: A technique that interleaves reasoning steps (“Thought”) with Actions and Observations to guide an LLM through planning and tool use.

ReAct: Tool-Calling

Tool example

```
from smolagents import tool
@tool
def calculator(expr: str) -> float:
    """
    Evaluate a basic arithmetic expression.

    Args:
        expr: The math expression to evaluate, such
              as "2 + 3 * (4 / 2)".
    """
    return eval(expr, {"__builtins__": None}, {})
```

High-level Concept

(Id) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: Search[Apple Remote]

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the Front Row media center program ...

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.

Act 2: Search[Front Row]

Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row [software]', ...]

Thought 3: Front Row is not found. I need to search Front Row [software] .

Act 3: Search[Front Row [software]]

Obs 3: Front Row is a discontinued media center software ...

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

Act 4: Finish[keyboard function keys]



ReAct loop

- Create a plan.
- Identify action using reasoning (CoT).
- Take action.
- Observe result and update plan.

Figure: <https://arxiv.org/abs/2210.03629>

ReAct: Early examples

You are an agent that answers questions by using two actions:

- search[query]: Google search the query. You can also use it to calculate math.
- finish[answer]: return the answer

Your generation should have the following format:

Thought: to analyze the observation and inform the action.

Action: your action

Question: If I have seven trillion dollars today, can I buy Apple, Nvidia, and Microsoft? If not, how much more money do I need?

Figure:

<https://rdi.berkeley.edu/llm-agents/f24>

Thought: I need to find the current market capitalization of Apple, Nvidia, and Microsoft to understand how much it would cost to buy all of them.

Action: search["current market capitalization of Apple, Nvidia, and Microsoft"]

Observation: "Nvidia's market cap stood at \$1.832 trillion as of about 10 a.m. ET Wednesday, while Alphabet's was at \$1.814 trillion, bringing Nvidia behind just Microsoft (MSFT) at \$3.015 trillion and Apple (AAPL) at \$2.836 trillion."

Thought: Now that I have the market cap for all companies, let's add them together and see if \$7 trillion is enough.

Action: search["1.832 trillion + 3.015 trillion + 2.836 trillion"]

Observation: 7683000000000

Thought: The amount of money needed in addition to the \$7 trillion is \$683 billion.

Action: finish["You need an additional \$683 billion to be able to buy Apple, Nvidia, and Microsoft."] 

Figure:

<https://rdi.berkeley.edu/llm-agents/f24>

Agency levels

Agency Level	Description	Short name	Example Code
☆☆☆	LLM output has no impact on program flow	Simple processor	<code>process_llm_output(llm_response)</code>
☆☆☆	LLM output controls an if/else switch	Router	<code>if llm_decision(): path_a() else: path_b()</code>
☆☆☆	LLM output controls function execution	Tool call	<code>run_function(llm_chosen_tool, llm_chosen_args)</code>
☆☆☆	LLM output controls iteration and program continuation	Multi-step Agent	<code>while llm_should_continue(): execute_next_step()</code>
☆☆☆	One agentic workflow can start another agentic workflow	Multi-Agent	<code>if llm_trigger(): execute_agent()</code>
☆☆☆	LLM acts in code, can define its own tools / start other agents	Code Agents	<code>def custom_tool(args): ...</code>

Section 3

LLMs in Code Development